

Laboratory Exercise No. 5

CRUD Application with Authentication Using LavaLust

Course: Web Systems and Technologies 2

Topic: CRUD Operations with Authentication

Framework: LavaLust

Database: Aiven MySQL

Deployment: Render

Objective

Create a simple CRUD web application using **LavaLust**. The application must connect to an **Aiven MySQL database** and be deployed on **Render**.

Instructions

1. Create the Database

Create a database in **Aiven MySQL**.

Create a table named:

products

Use the following columns:

Column	Data Type	Description
id	INT	Primary Key, Auto Increment
product_name	VARCHAR(100)	Product name
description	TEXT	Product description
price	DECIMAL(10,2)	Product price
quantity	INT	Available quantity
created_at	TIMESTAMP	Date created

2. Create the LavaLust Application

Using LavaLust, create:

ProductModel

ProductController

Product views

The application must implement the following CRUD operations:

- **Create** – Add a product
- **Read** – Display all products
- **Update** – Edit a product
- **Delete** – Remove a product

3. Apply Authentication

Protect the product management pages using **LavaLust Middleware** or any appropriate authentication mechanism.

Only authenticated users should be allowed to access:

/products

/products/create

/products/edit/{id}

/products/delete/{id}

Students may use:

- LavaLust middleware
- Session-based authentication
- Any other appropriate authentication mechanism

The objective is to ensure that **unauthenticated users cannot directly access the CRUD pages**.

4. Connect to Aiven

Configure the LavaLust application to connect to the **Aiven MySQL database**.

Students must use the database credentials provided by Aiven.

Do not commit database passwords or other sensitive credentials to GitHub.

5. Deploy to Render

Push the completed LavaLust project to GitHub and deploy it to **Render**.

Configure the required database credentials using **Render Environment Variables**.

6. Test the Application

The deployed application must demonstrate:

Login

↓

Authenticated User

↓

Product Management

├— Create

├— Read

├— Update

└— Delete

Students must verify that:

- An unauthenticated user cannot access the product management pages.
- An authenticated user can perform all CRUD operations.
- Data is stored in and retrieved from Aiven MySQL.
- The application works through the Render URL.

Submission

Submit:

1. **GitHub Repository URL**
2. **Render Application URL**
3. Screenshots showing:
 - Login/authentication
 - Product list
 - Add product
 - Edit product
 - Delete operation
4. Screenshot of the products table in Aiven
5. Working CRUD application